Dikshant Shehmar

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Education

University of Alberta (*UofA*) — *Edmonton, Canada*
PhD in Computing Science — *(May '26 - Present)*
Supervisor: Marlos C. Machado
Research Interests: Reinforcement Learning, Continual Learning, Autonomous Driving and Robotics

University of Alberta (*UofA*) — *Edmonton, Canada*
Masters in Computing Science — *(Aug '24 - May '26)*
Supervisors: Matthew Taylor | Marlos C. Machado

Indian Institute of Technology Bombay (*IITB*) — *Mumbai, India*
Bachelors of Technology in Civil Engineering, Dept. Rank: 5/112 — *(Jul '18 - May '22)*
With Minor in *Electrical Engineering*

Masters' Thesis

Laplacian Representations for Decision-Time Planning — *(May '25 - May '26)*

My research focuses on representation learning and planning in reinforcement learning, with a focus on Laplacian representations. We proposed a hierarchical planning algorithm that exploits the Laplacian structure for subgoal identification and distance estimation, showing strong results on goal-conditioned navigation and manipulation tasks.

Research Experience

An Adaptive Transition Framework for Game-Theoretic Based Takeover — *(Oct '24 - Feb '25)*
Advisors: Prof. Matthew Taylor and Prof. Ehsan Hashemi — *University of Alberta, Canada*

- Proposed an adaptive game-theoretic framework for shared humanautomation control during vehicle takeovers
- Demonstrated a **14.4%** improvement in trajectory accuracy via real-time adaptive control in ISO-standard simulations
- Modeled driver-specific behavior within shared control systems using CARLA-based human-in-the-loop experiments

Scalable Multi-Agent RL Training School for Autonomous Driving - SMARTS — *(May '21 - Aug '21)*
Advisor: Prof. Matthew Taylor, Intelligent Robot Learning Lab — *University of Alberta, Canada*

- Constructed a training pipeline in SMARTS for multiple ego agents across varying map and traffic-pattern complexities
- Trained single ego-agent scenarios using **PPO** over increasingly difficult maps and traffic, **modelling human learning**
- Initialised multiple ego-agent training with single-agent weights, obtaining **3x faster convergence** over a fixed horizon
- Enhanced T-intersection safety by training agents with additional **hand-coded** positional data shared across neighbors

Multi-Agent Patrolling — *(Apr '20 - Apr '21)*
Advisors: Prof. Arpita Sinha and Prof. Leena Vacchani, Department of Systems & Control — *IIT Bombay*

- Devised an **IoT**-based multi-robot patrolling algorithm using **DQN** for effective monitoring in remote areas
- Created an interface for communication between agents in **SUMO** using Traffic Control Interface (**TraCI**) library
- Formulated a **decentralized junction-based** decision making approach, reducing junction visit intervals by **2x**
- Co-authored a research **manuscript** titled "*Interconnecting Vehicles using IoT Framework for Multi-Agent Patrolling*"

SeDriCa | Unmesh Mashruwala Innovation Cell — *(Aug '20 - Apr '22)*
*Worked on a **20+** member team aiming to develop a **Se**lf-**Dri**ving **Ca**r capable of traversing unstructured environments*

- Designed a hybrid **MPC** controller using **dynamic bicycle model** to attain higher speeds and tackle adverse scenarios
- Crafted a drive-by-wire (**DBW**) system from ground up for mapping accelerator pedal position to DBW throttle body
- Implemented RRT* and Hybrid A* for path planning and optimized the route by integrating data from Google Maps

Publications and Preprints

P1. Shehmar, D., Schlegel, M., Taylor, M. E., & Machado, M. C. *Laplacian Representations for Decision-Time Planning*. In International Conference on Machine Leaning (ICML), 2026. - `arXiv:2602.05031`

P2. Shehmar, D., Taylor, M., & Hashemi, E. *An Adaptive Transition Framework for Game-Theoretic Based Takeover*. *Preprint*, 2025. - `arXiv:2510.10893`

Work Experience

Data Driven Process Control *(Nov '24 - Feb '25)*
Advisor: Payam Mousavi, Alberta Machine Intelligence Institute (AMII) *AMII | Repsol Spain*

- Built data-driven nonlinear control models predicting industrial chemical processes without first-principles equations
- Benchmarked advanced control strategies including RL algorithms (PPO, SAC), DMDc, NMPC and PID
- Optimized distillation column and CSTR performance through comparative analysis of multiple control approaches

Infrastructure Cooperative Safety Assist System *(Oct '22 - Aug '24)*
Advisor: Mr. Tokitomo Ariyoshi, Chief Engineer, CS Domain *Honda R&D Ltd. Japan*

- Spearheaded project to enhance safety of Honda's self-driving car in **occluded regions** using infrastructure cameras
- Modeled traffic-user trajectories with intelligent driver model (**IDM**) implemented using Frenet Coordinates system
- Optimized (**3x faster**) and deployed Collision Risk Map algorithm to proactively alert VRUs to potential collisions
- Implemented **3D-Net** model for monocular object detection to obtain position and velocity of traffic participants
- Established remote control room to access live CCTV streams utilizing python socket library over **4G** networks
- Simulated collision detection on Gazebo and employed advanced **HMI** techniques to alert users of potential collisions

Camera and Radar Fusion *(May '22 - Jul '22)*
Advisor: Mrs. Misa Komuro, Assistant Chief Engineer, CS Domain *Honda R&D Ltd. Japan*

- Merged camera and radar-based object detection results to improve traffic user detection efficiency by **25%**
- Applied temporal and spatial alignment techniques to **synchronize** sensor inputs, increasing tracking precision
- Visualized fusion results on **RViz** using ROS Visual Markers and tested robustness of algorithm in crowded regions

Room Service Automation Bot *(Jun '20 - Jul '20)*
Golden Oak Projects Private Limited *Tech startup*

- Developed an autonomous hotel service robot adept at delivering amenities and responding to room service orders
- Implemented **SLAM** for real-time mapping of environment by integrating data from encoder, IMU and 2D-Lidar
- Simulated scenarios on custom-made Gazebo environment and utilized **DWA** algorithm for efficient navigation

Scholarships and Awards

- Awarded **University of Alberta Graduate Recruitment Scholarship** (Value: *$5000 CAD*) *('25)*
- Acquired **N5** level of certification in the Japanese Language Proficiency Test (**JLPT**) *('23)*
- Conferred with **Department Technical Excellence Award**, given annually to **1** in **400** students *('22)*
- Recipient of Kishore Vaigyanik Protsahan Yojana (**KVPY**) Fellowship awarded by Government of India *('18)*
- Secured **97.8 percentile** in JEE Advanced entrance examination among **0.15** million+ candidates *('18)*

Reviewer

- Neural Information Processing Systems (**NeurIPS**) 2026
- International Conference on Learning Representations (**ICLR**) 2026
- Reinforcement Learning Conference (**RLC**) 2025, 2026
- European Workshop on Reinforcement Learning (**EWRL**) 2026

Teaching Assistant

CMPUT 365: Introduction to Reinforcement Learning *(Fall '25)*
Department of Computing Science, University of Alberta

CMPUT 503: Experimental Mobile Robotics *(Spring '25)*
Department of Computing Science, University of Alberta

CMPUT 366: Search Planning in Artificial Intelligence *(Fall '24)*
Department of Computing Science, University of Alberta

MA108: Differential Equations *(Spring '22)*
Department of Mathematics, Indian Institute of Technology Bombay

Technical Projects

Intelligent Picking Robot | Flipkart Grid 2.0-Robotics Challenge *(Jun '20 - Aug '20)*
Led a team of 5 developing a robotic arm capable of picking and transporting items in a warehouse | Top 2% across India

- Designed a **4-DoF** robotic manipulator and visualized its grasping mechanism on **RViz** using **MoveIt** framework

- Implemented **YOLOv3** algorithm for object detection and identified potential grasping points using **OpenCV** functions
- Mapped area between pick and drop stations using **SLAM** and employed **A*** algorithm for efficient path planning

Terrace Farming Bot | Inter-IIT Technical Meet 8.0 *(Oct '19 - Dec '19)*
Worked in a team of 8 to develop an autonomous step-climbing robot to perform terrace farming operations

- Employed a linear actuator mechanism using stepper motors and developed **CAD** models for precise manufacturing
- Fused monocular **visual odometry** and data from MPUs and stepper motor encoders for efficient state estimation
- Deployed a **PID**-based position and orientation controller employing ultrasonic sensors to provide accurate feedback

Parallelizing A* and D* Algorithms | High Performance Scientific Computing *(Feb '21 - May '21)*
Advisor: Prof. S. Gopalakrishnan, Mechanical Engineering Department *Course Project*

- Enhanced computational efficiency of A* and D* by parallelizing them with **OpenMP** and **CUDA** frameworks
- Demonstrated and visualised improved path planning of parallel A* by implementing it using **pygame** library
- Benchmarked the **Serial**, **OpenMP**, and **CUDA** (for GPUs) implementations of A* and D* algorithms

Autonomous Quadruped Robot | RoboCup Rescue League Challenge *(Sep '19 - May '20)*
*Part of **17** member team working on autonomous quadruped for easy maneuvering on any terrain*

- Performed **Inverse Kinematic** analysis for quadruped control to ensure optimum balance on uneven ground
- Studied rhythmic movement of each leg of quadruped for optimal **gait selection** based on speed and terrain

Other Technical Activities

- Delivered **talks** on RL for autonomous driving and Control Theory to **70+ students** with demonstrations *('20)*
- Guided **150+** freshmen teams in XLR8 competition to build and debug wireless bluetooth controlled bot *('19)*
- Replicated **Thor's Hammer** using electromagnetism and RFID for Institute Technical Council Orientation *('19)*
- Designed a **line-follower robot** capable of solving complex maze in Institute's technical fest competition *('19)*
- Assembled a **bluetooth-controlled robot** and applied differential-drive mechanism for optimum control *('19)*
- Developed **Xylobands** to represent robotics club in Institute Technical Orientation for freshmen batch *('19)*

Technical Proficiency

Languages	Python, C++, Matlab, OpenCL, Octave, Bash, CSS, Markdown, LaTeX
Softwares	Webots, Gazebo, CARLA, SUMO, OSM, Netedit, TraCI, Simulink, Auto-CAD
Frameworks/Libraries	JAX, PyTorch, Tensorflow, RLlib, ROS, Gym, Sockets, Optax, CV, Pygame

Leadership and Mentorship Roles

Team Leader | SeDriCa, UMIC *(May '21 - May '22)*
Led a 26-member team representing IIT Bombay at the International Ground Vehicle Competition

- Managed operations, finances, logistics and knowledge-transfer in a **4-tier**, **multi-disciplinary** student technical team
- Raised **Rs 3.5 M** from IITB and Mahindra RISE, and forged relations with multiple academic and industry experts

Technical Head | Inter-IIT Tech Meet 9.0 *(Jan '21 - Mar '21)*
Part of a 50-member strong contingent working on 10 projects of medium and high-level difficulties

- Conducted **40+** stringent interviews for selecting contingent managers, project leads, sponsors and financial advisors
- Provided critical technical support for resolving issues in docker, ROS, gazebo, and integration in various projects

Convener | Electronic and Robotics Club, Institute Technical Council *(May '19 - Apr '20)*
Part of a 15+ member team that conceptualizes and organizes events for tech enthusiasts in the Institute

- Hosted a widely-attended **ROS workshop** aimed at beginners during COVID, drawing over **300 tech enthusiasts**
- Spearheaded XLR8, handling **600+** participants (up **30%**), achieving highest bot-completion rate (**92%**) in four years

Key Courses Undertaken

Computer Science	Deep Reinforcement Learning, Optimization and Decision Making under Uncertainty, Artificial Intelligence and Machine Learning, Deep Learning Specialization
Controls and Robotics	Autonomous Driving Motion Planning & Coordination of Autonomous Vehicles, Adaptive Control theory, Intelligent Feedback & control systems

EXTRACURRICULARS

Sports

- Represented Computer Science Graduate Student Association (CSGSA) of UofA in winter Volleyball Intramurals, 2024 and summer Soccer Intramurals, 2025
- Secured **1st place** in Institute's annual inter-hostel football & cricket tournaments 2021
- Represented IITB in Football, Cricket and Athletics at **Annual Training Camp** 2019

Volunteering

- Served as an **NCC cadet** in 2-Maharashtra Engineering Regiment, 2018-19 contingent
- Selected as cadet for **Republic Day Parade** '19; awarded with NCC ATC certificate
- Counseled students from local municipal schools for IITB's **Career Counseling Campaign**

Miscellaneous

- Bagged **1st position** in **Logic GC** organized by Maths and Physics Club of IITB
- Awarded **Hostel Technical Special Mention** for exemplary contribution to institute tech